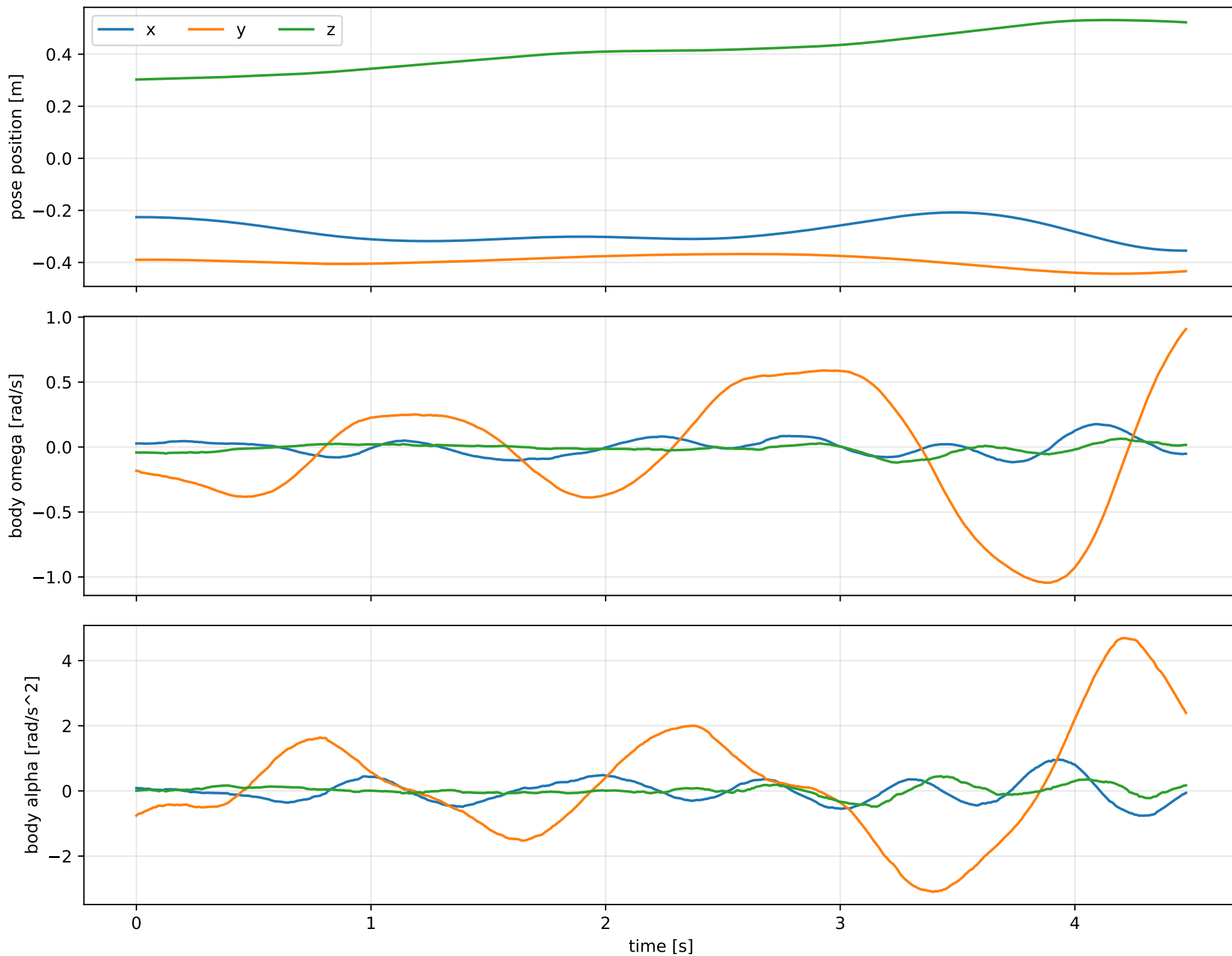
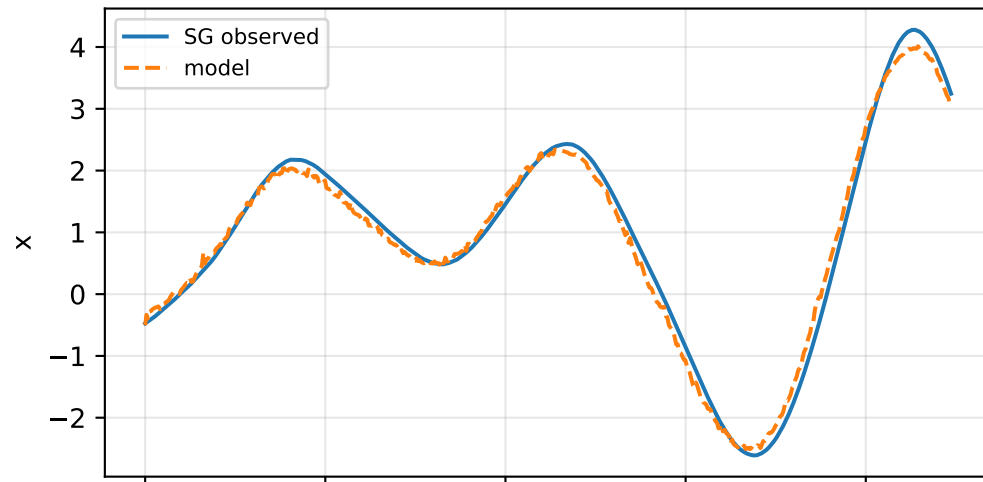


gimbal_command_replay: SG trajectory and derived motion

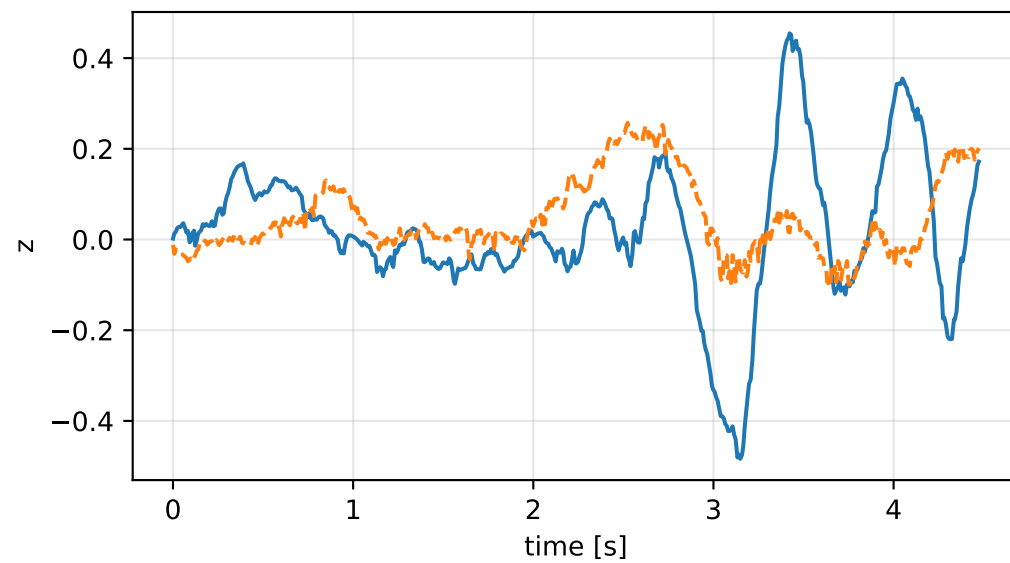
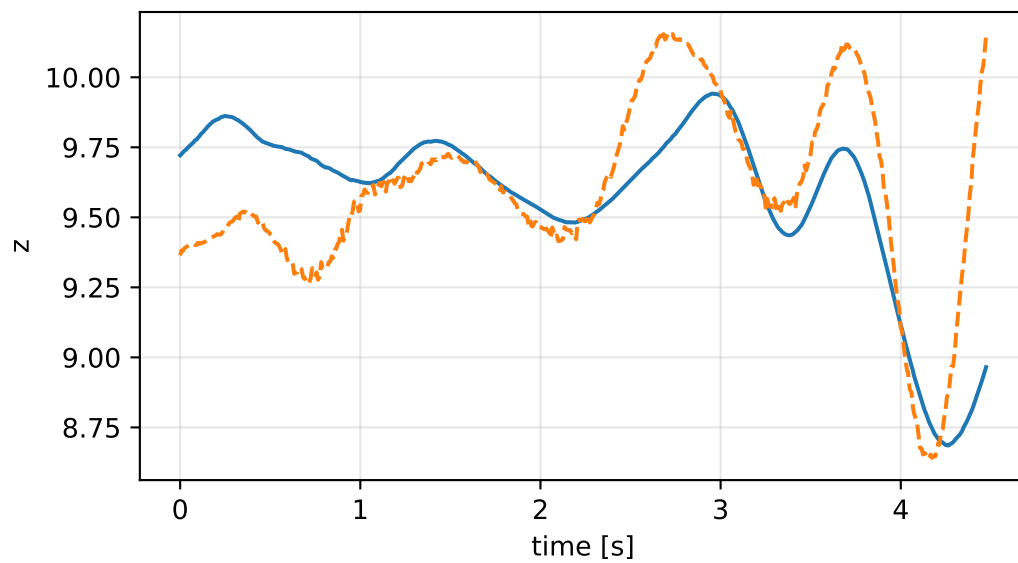
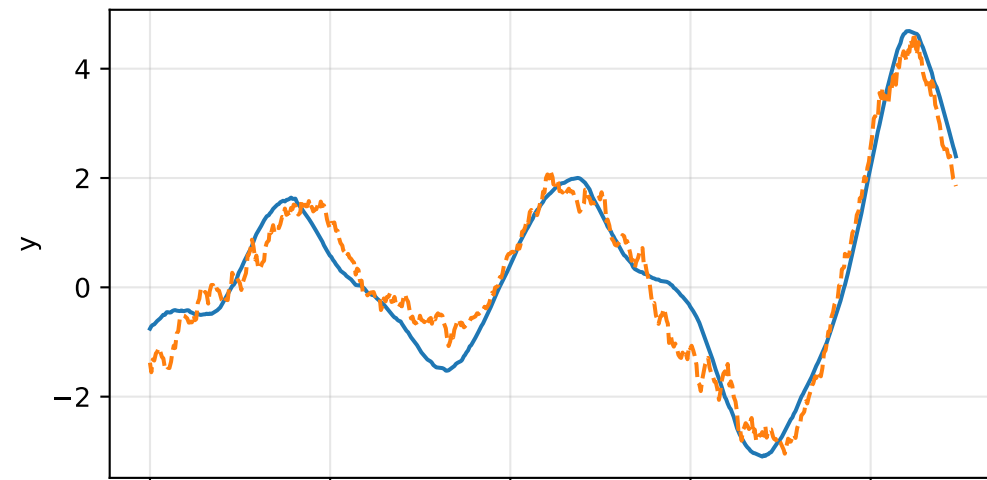
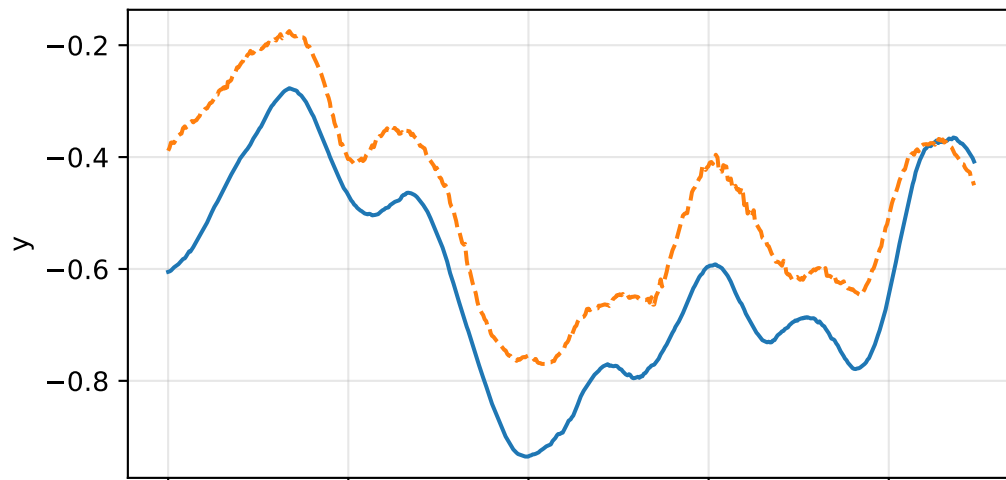
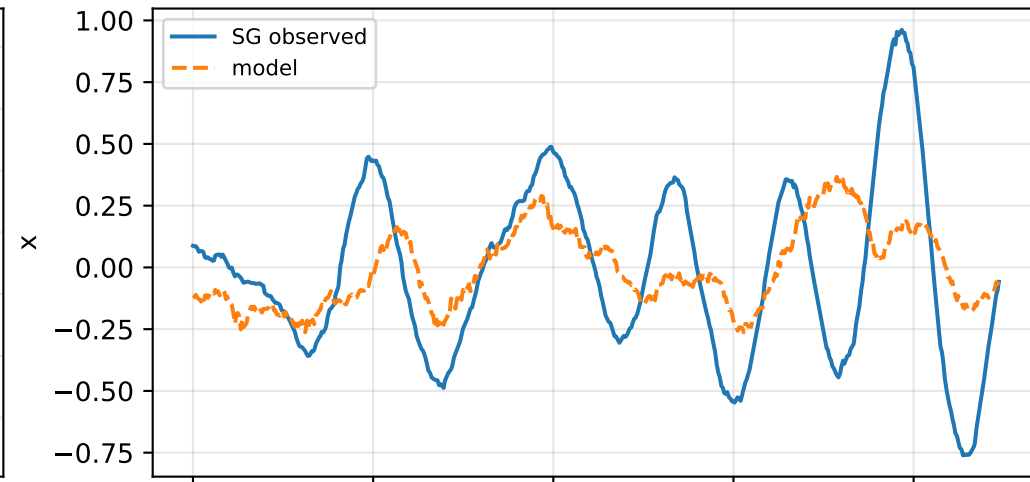


gimbal_command_replay: acceleration residual objective

specific acceleration [m/s^2]

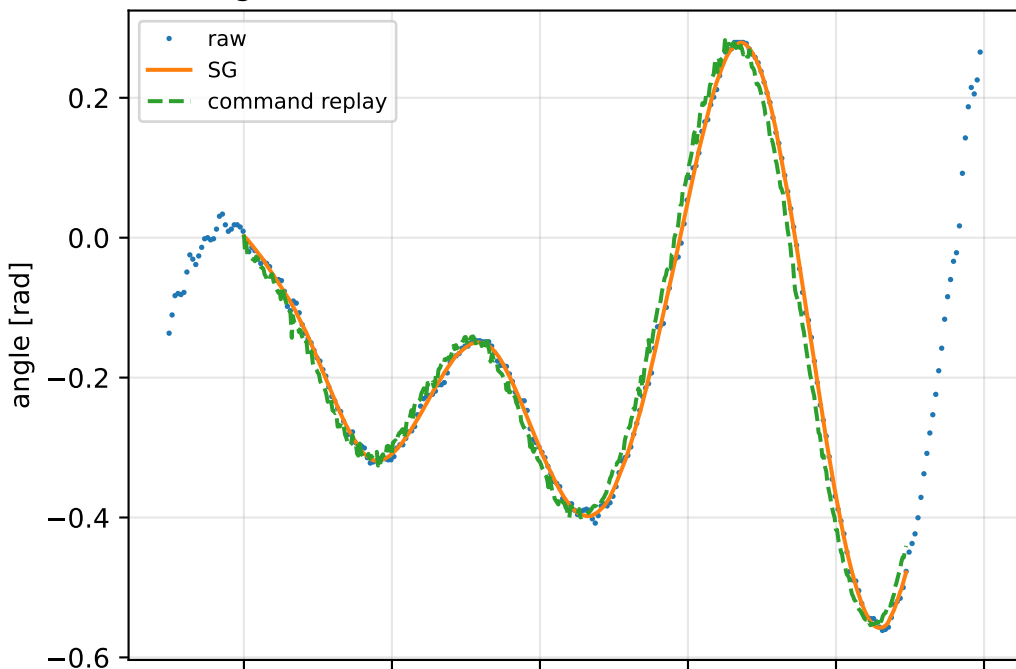


angular acceleration [rad/s^2]

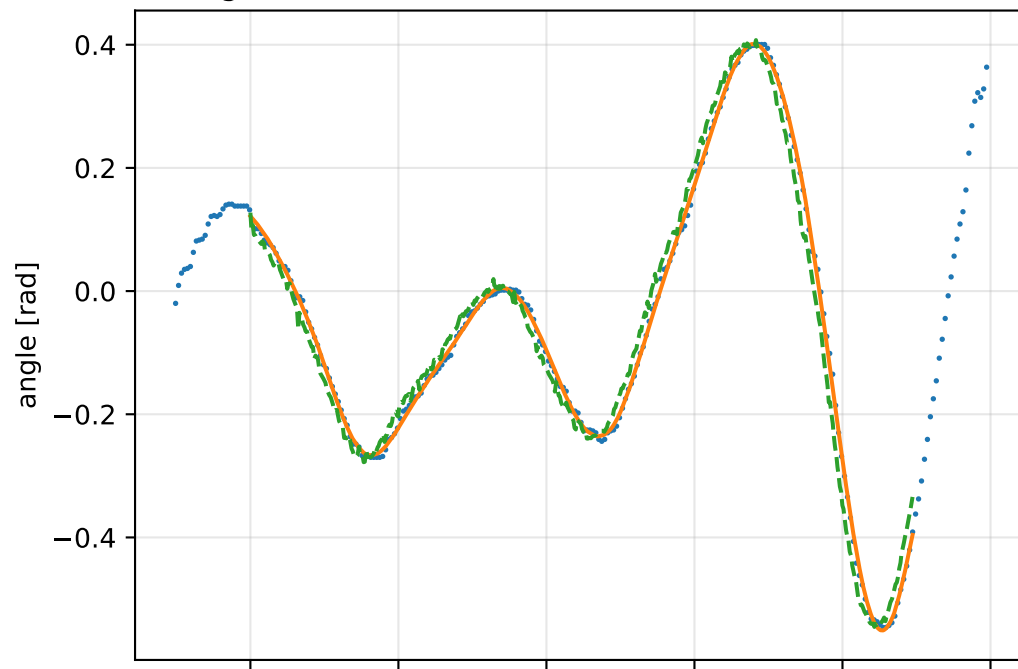


gimbal_command_replay: gimbal measurement audit (objective=command_replay)

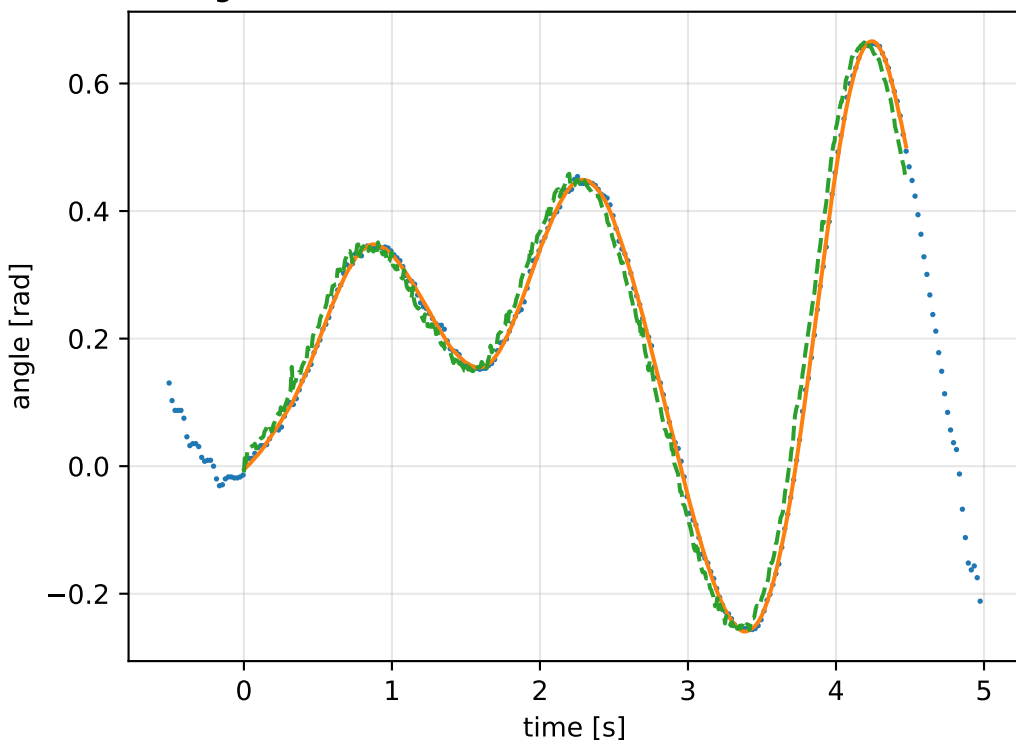
gimbal 1: RMSE=0.027, max=0.07529 rad



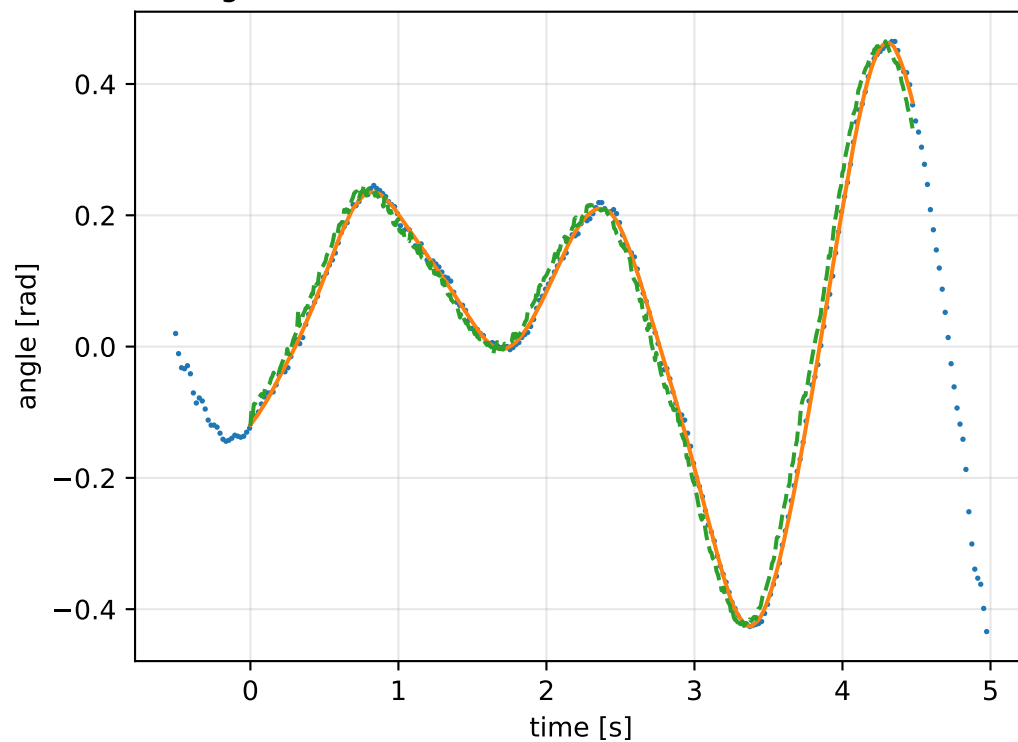
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

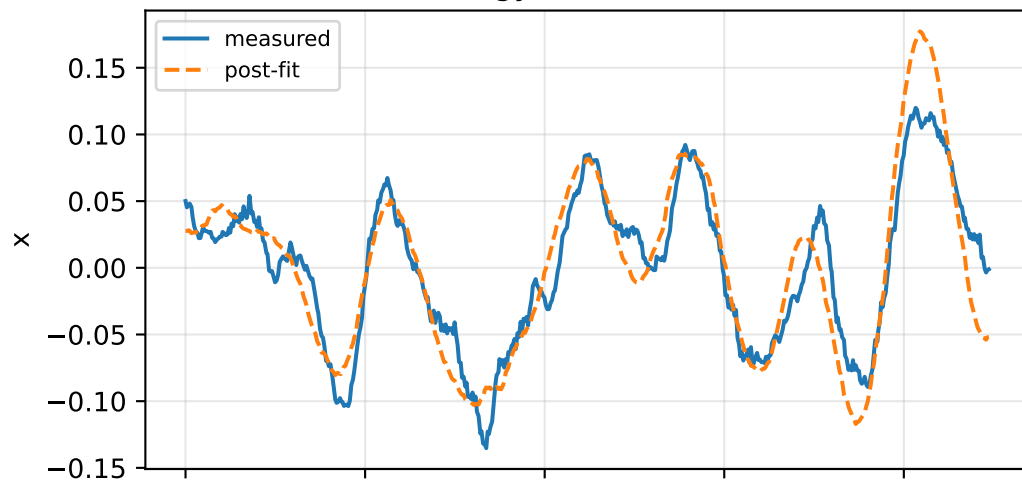


gimbal 4: RMSE=0.02598, max=0.07261 rad

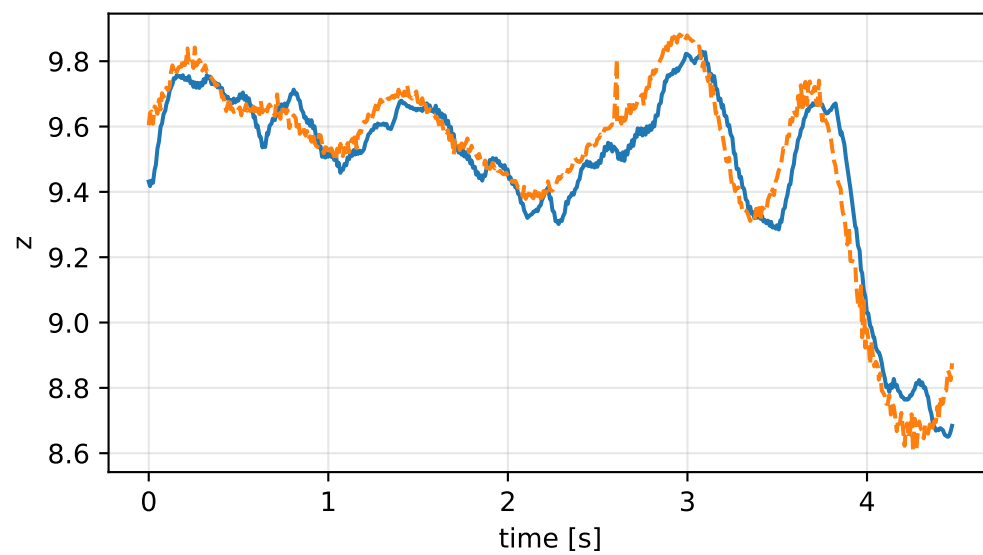
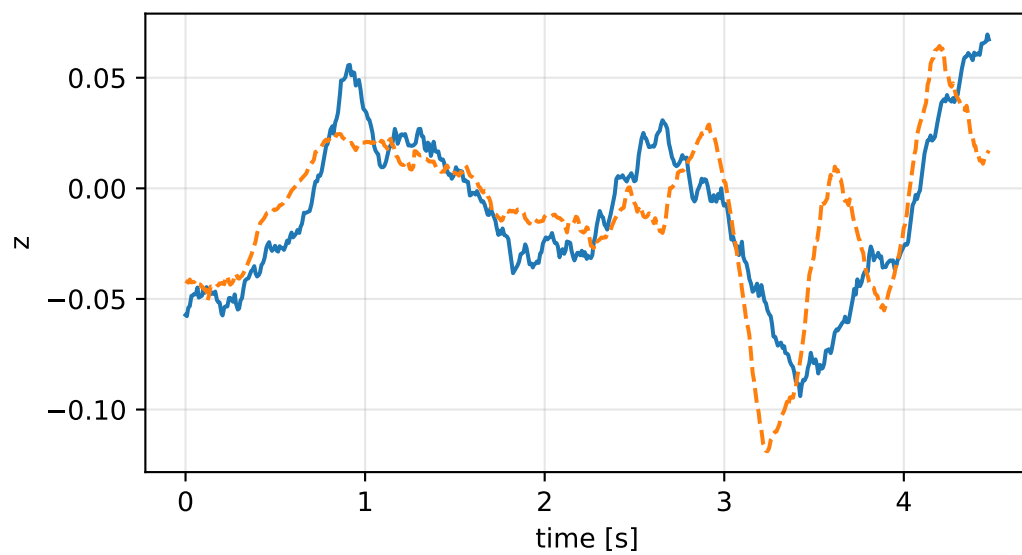
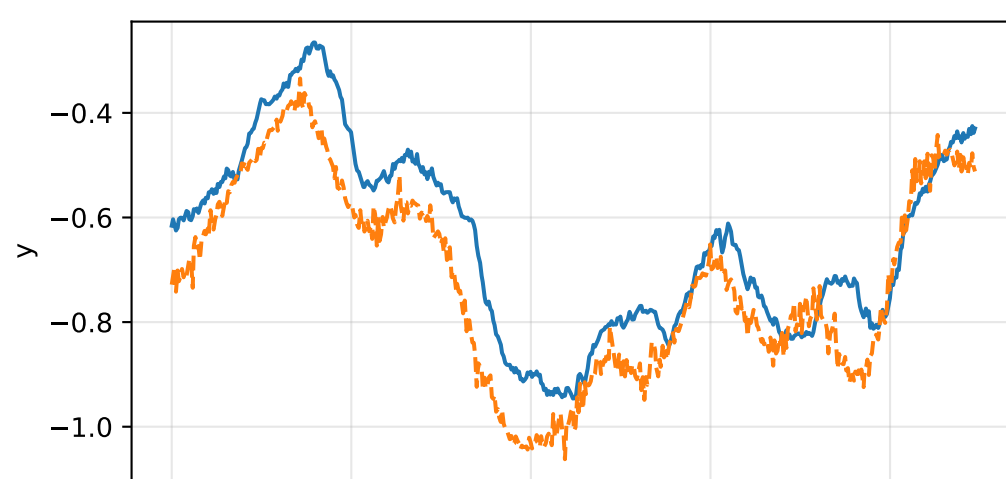
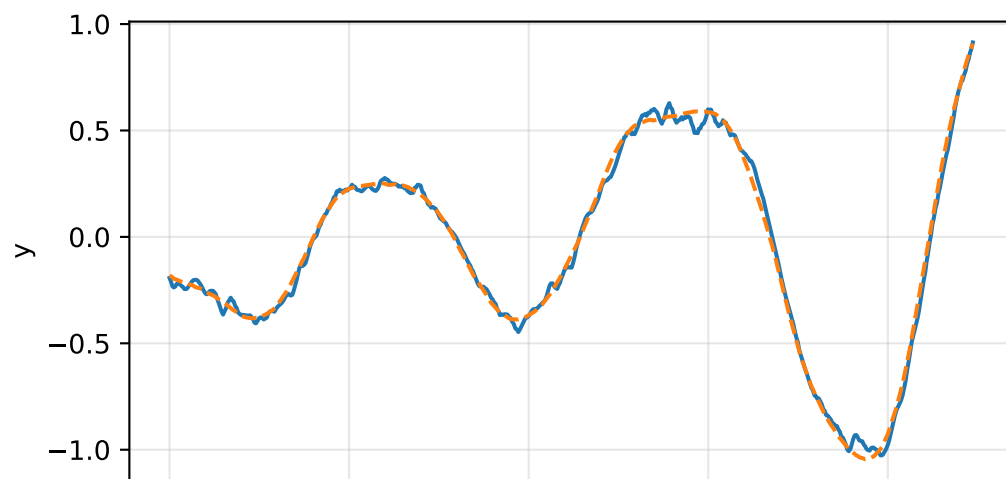
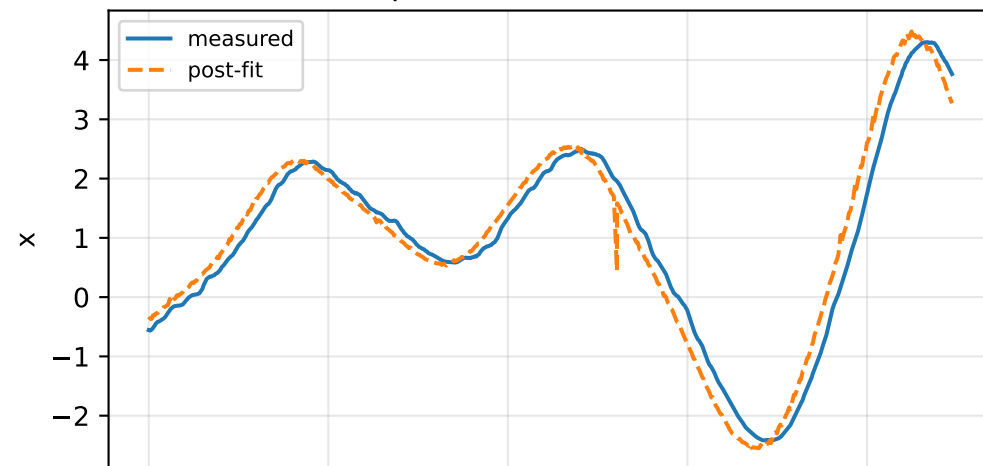


gimbal_command_replay: IMU comparison (diagnostic only)

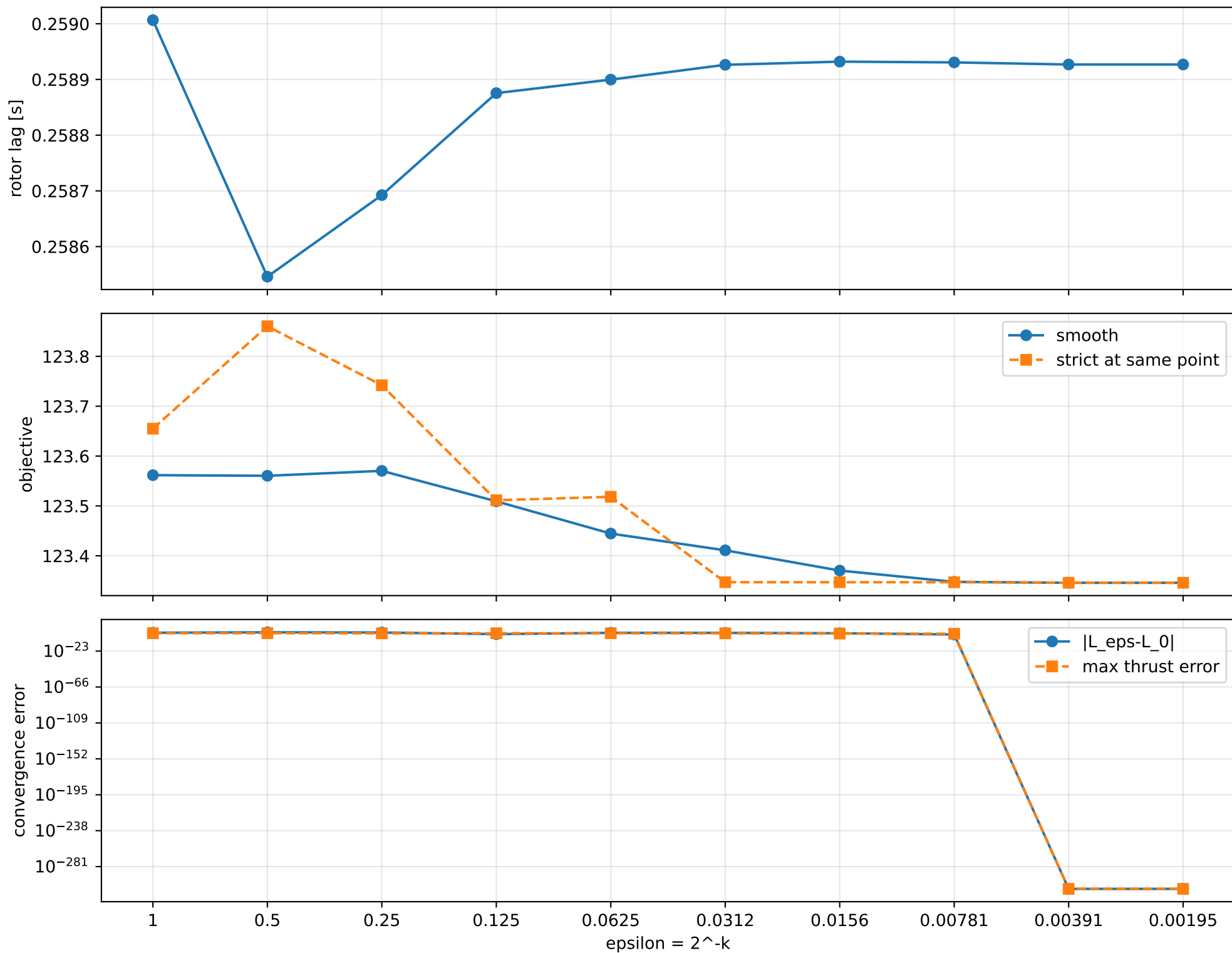
gyro [rad/s]



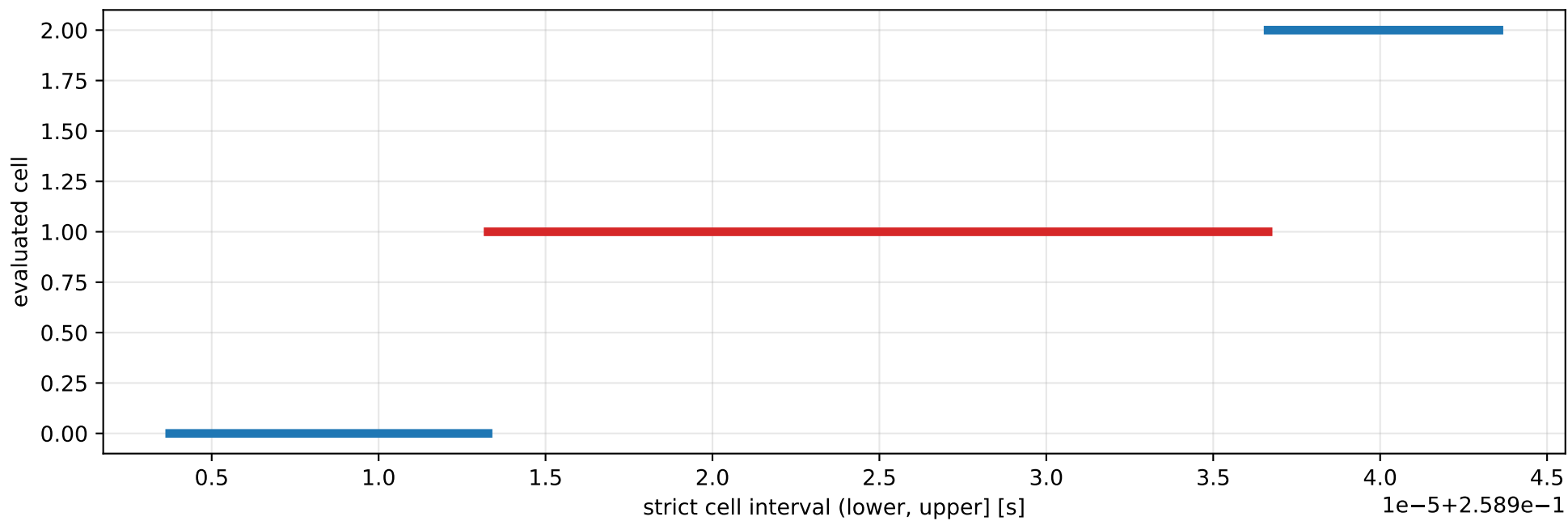
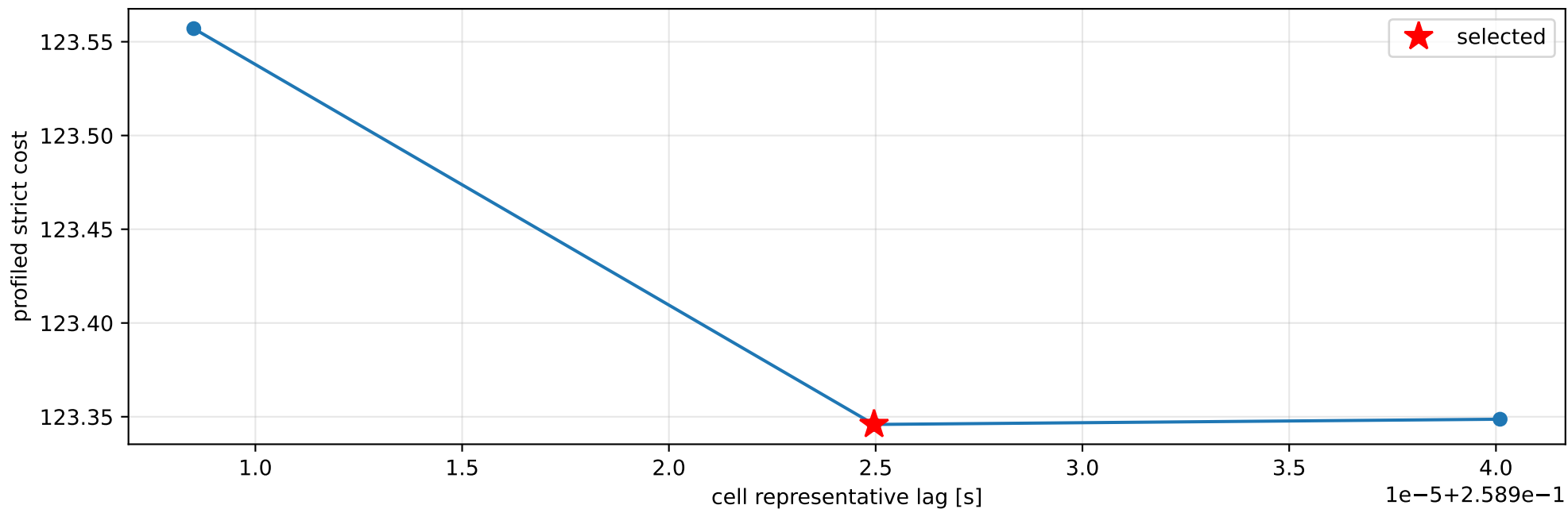
specific force [m/s^2]



gimbal_command_replay: smooth-to-strict continuation



gimbal_command_replay: exact strict-ZOH lag cells



selected (0.258913279, 0.258936644] s; final smooth 0.258926873 s; data boundary=False

gimbal_command_replay: parameter estimate

Case: gimbal_command_replay

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 4.97182701

inertia [kg m²]:

```
[[ 0.32752704  0.01175763 -0.04184988]
 [ 0.01175763  0.15993295 -0.00984825]
 [-0.04184988 -0.00984825  0.46260202]]
```

CoG body [m]: [-0.02968859 0.00241618 0.01124373]

force effectiveness: [1.96658586 1.60742019 1.5672905 2.01967727]

scale-free J/m [m²]:

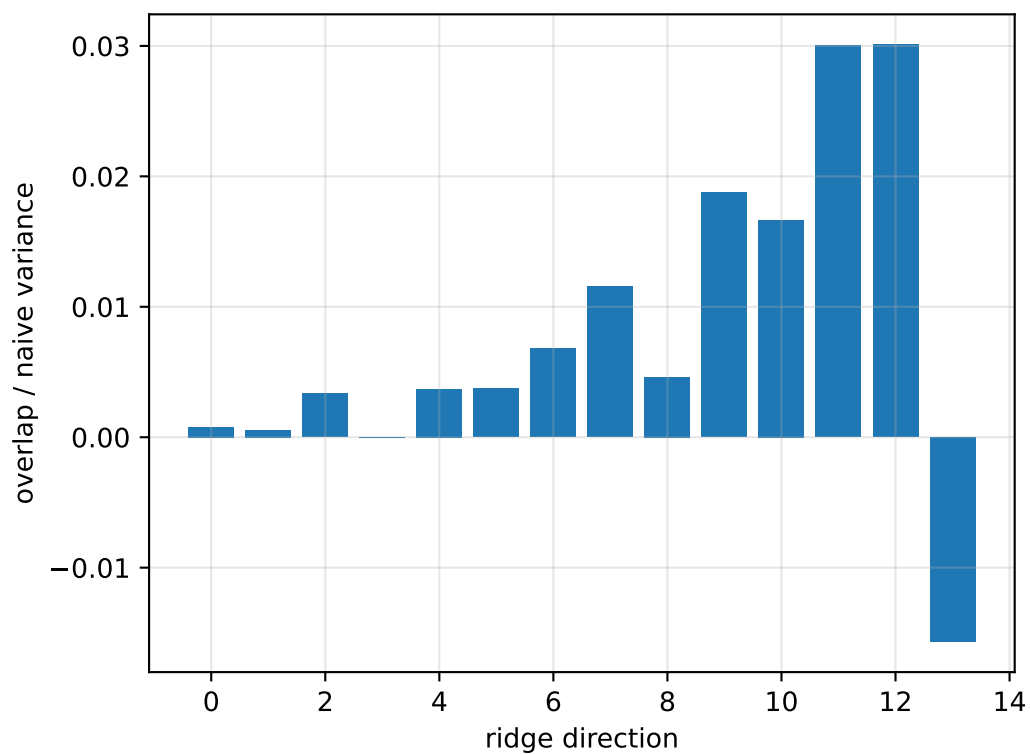
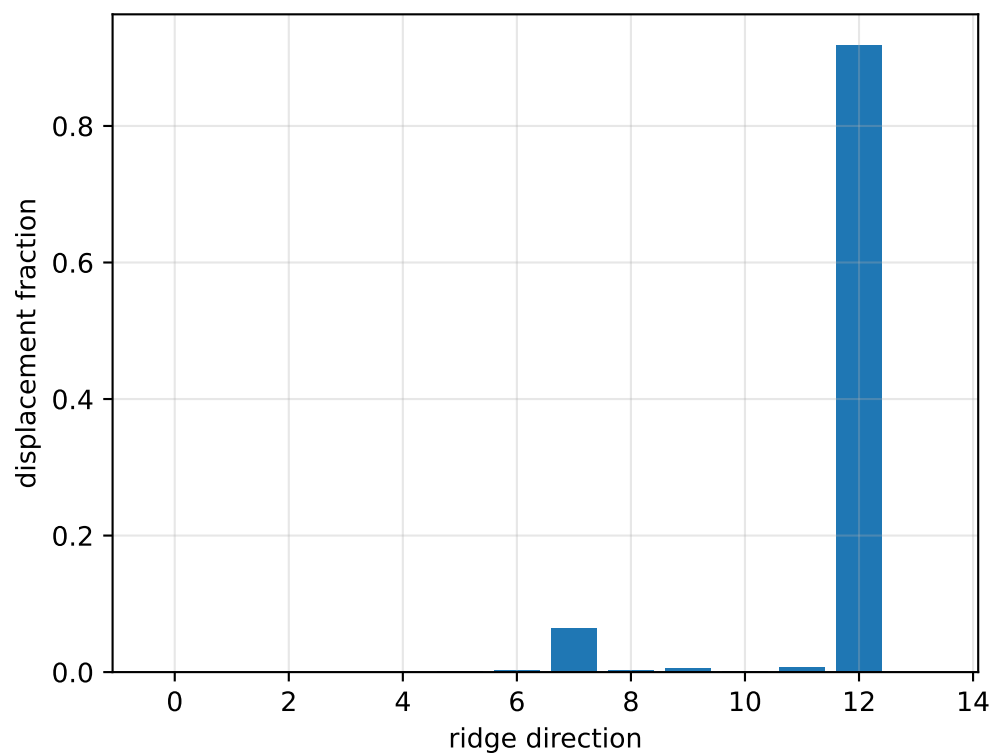
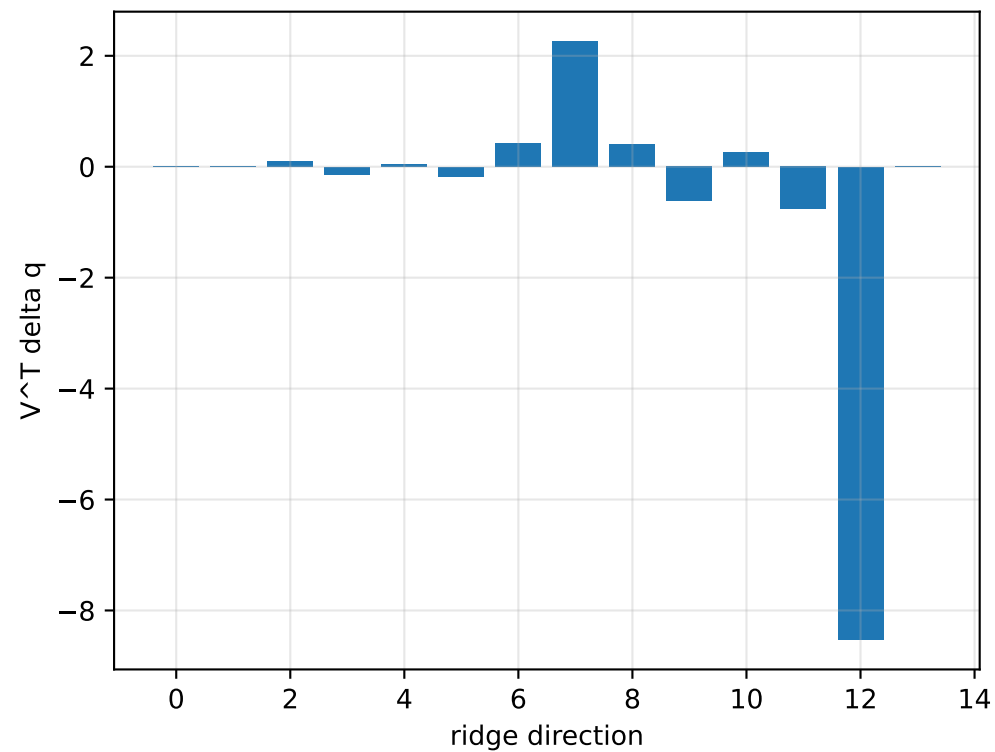
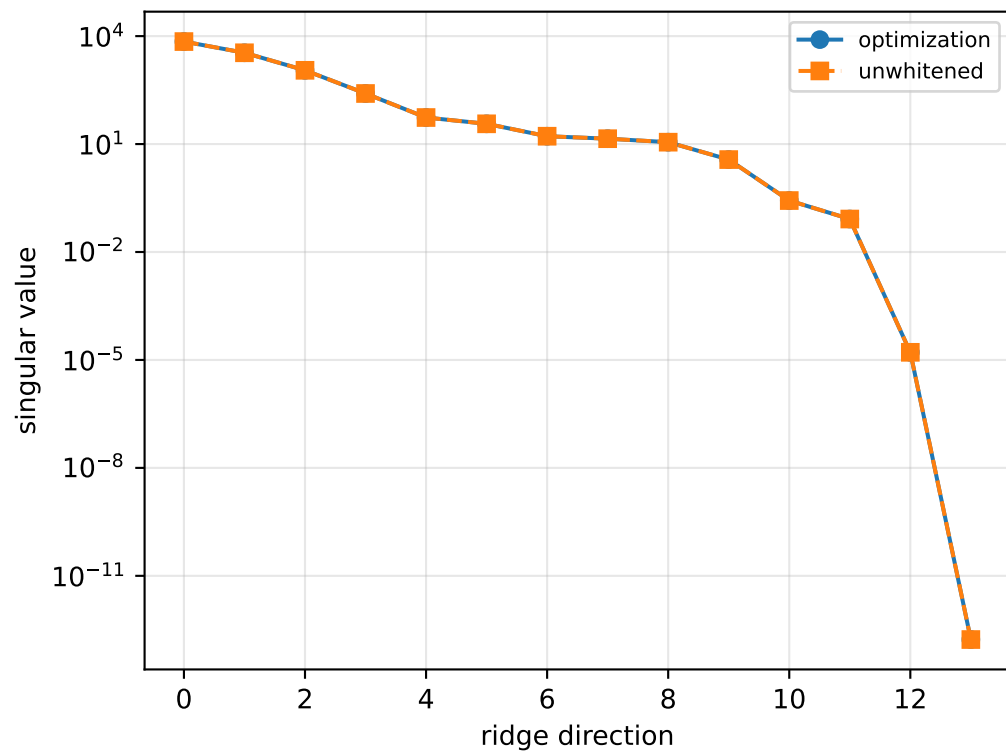
```
[[ 0.0658766  0.00236485 -0.0084174 ]
 [ 0.00236485  0.03216784 -0.00198081]
 [-0.0084174 -0.00198081  0.09304467]]
```

scale-free f/m: [0.39554591 0.32330574 0.31523432 0.40622437]

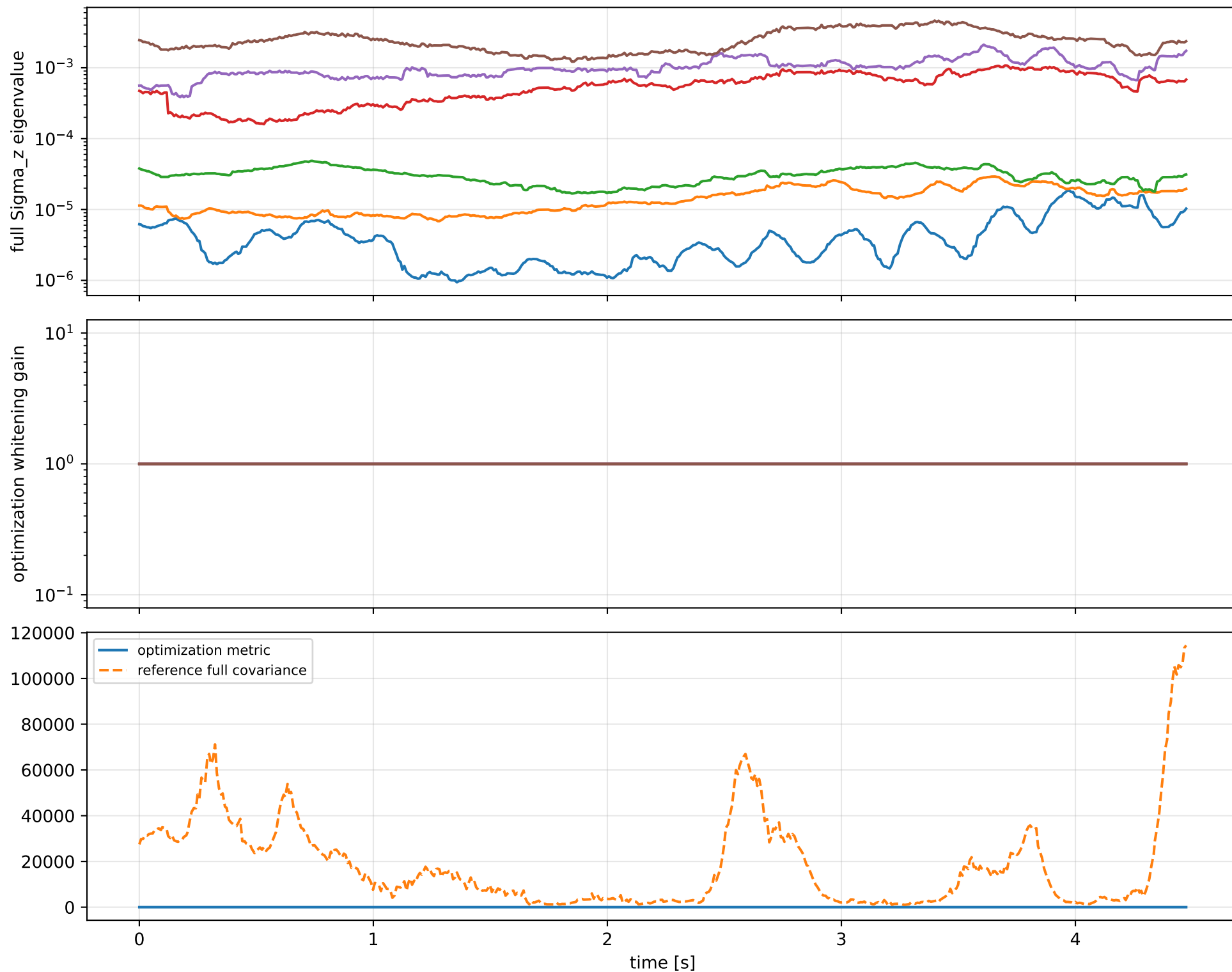
rotor lag cell: (0.258913279, 0.258936644] s

representative: 0.258924961 s

gimbal_command_replay: ridge spectrum (rank 13, nullity 1, $||v||=1.6e-13$)



gimbal_command_replay: optimization vs reference covariance



gimbal_command_replay: wrench / closure (force max 1.27e-14, torque max 2.01e-16)

